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1. VC Returns & Performance Persistence

How do VC returns compare to public markets?

VC has a higher average return than public equities, but the more important difference is **dispersion** — the gap between top and bottom performers is enormous (roughly 18% median-to-top-5th-percentile vs. ~2% for long-only public strategies). This means picking the right VC fund matters far more than picking the right equity manager.

Performance persistence — unique to VC

In most asset classes, strong past performance does not predict future performance — managers revert to the mean. VC is a striking exception: **top-quartile funds are significantly more likely to remain top-quartile in their next fund**. Bottom-quartile funds also tend to stay bottom-quartile.

Quartile	Persistence Pattern
Previous fund quartile 1 (top)	45% chance of being Q1 again; avg IRR 35.6%, MOIC 4.1x
Previous fund quartile 4 (bottom)	44% chance of staying Q4; avg IRR 2.6%, MOIC 1.3x
First-time funds	Fairly even distribution; avg IRR 15.6%, MOIC 2.5x — but wide spread

Why does persistence exist?

Unlike public markets, VC returns depend heavily on proprietary deal access, founder relationships, and the ability to attract top companies to your portfolio. These are durable competitive advantages that compound over time. A top-quartile fund's brand attracts better deals, which generates better returns, which reinforces the brand — a virtuous cycle.

2. Sources of High VC Returns

Why do some VCs consistently outperform? The lecture identifies four main levers:

Source	Detail
Investing style	Sector, stage, and geography choices drive returns over certain periods. The cleantech wave (2006–2011) is a cautionary tale: nearly every VC wanted exposure, but long development timelines and high capital intensity made the asset class a poor fit for VC fund structures.
Deal sourcing & selection	Some VCs are simply better at identifying promising companies. Voting structure in investment committees matters: a <i>champion model</i> (one partner can push through a deal) funds more experiments and catches more outliers but also more duds; a <i>unanimous model</i> misses more winners. Neither is universally optimal.
Access to competitive deals	VC is a matching market — entrepreneurs choose their investors too. High-reputation VCs get into oversubscribed rounds that others cannot access. Affiliation with a top-tier firm is itself a signal entrepreneurs pay for (via lower valuations or better terms).
Post-investment management	Governance, board seats, and operational value-add all matter. But the <i>most</i> powerful post-investment tool is effective management of multi-stage financing — using abandonment options and follow-on investment strategically.

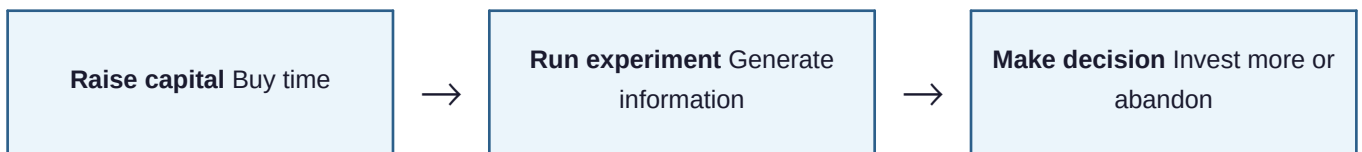
The 'hits' insight

~60% of VC investments return 1x or less. But the >10x bucket — just 5% of investments — generates over 50% of total gross returns. The entire VC model depends on a few outsized winners. This is why VCs must **keep funding promising companies aggressively** even under uncertainty, and why early selection of companies with genuine home-run potential matters more than average quality.

3. Multi-Stage Financing as Real Options

The core logic

Every round of financing should be understood as **buying time to run an experiment**. The experiment generates information. That information enables a better decision: invest more (exercise the option) or walk away (let the option expire). This is the real options framework applied to venture investing.



Experiments that cost less and reveal more information are enormously valuable — because they let you put **less money into losers** and **more money into winners**. This is the fundamental insight behind staged financing.

4. The Staged Bet Example (Panels A & B)

Both panels below have the same **unconditional 1% chance of a £1B payoff**. But they produce very different decisions:

PANEL A — All-or-Nothing Bet

Invest £11M today for a 1% chance of £1B (99% chance of £0).

Expected value = $1\% \times £1B = £10M < £11M$ cost → NPV negative. Do not invest.

PANEL B — Staged Financing

Same 1% unconditional success probability, but split into two stages:

- **Stage 1:** Run a cheap experiment. It succeeds with 10% probability.
- **Stage 2:** If the experiment succeeds, invest £11M for a 10% chance at £1B.

If it fails, walk away — spending nothing further.

Working backwards from Stage 2:

If you reach Stage 2, $EV = 10\% \times £1B = £100M$ post-money. You invest £11M, pre-money = £89M. Since $EV > \text{cost}$, you invest if the experiment succeeds.

Value of the experiment (Stage 1):

= $10\% \text{ chance} \times £89M \text{ pre-money value} = £8.9M$

You would pay up to £8.9M for the experiment alone — nearly as much as the full £11M all-or-nothing bet, but you only deploy the big capital when prospects are good.

The key insight

It is NOT the odds that change between Panel A and B — both have a 1% unconditional success rate. What changes is **when you commit capital**. Staging lets you avoid spending £11M on the 90% of experiments that fail. The option to abandon is real and enormously valuable.

5. When Real Options Are Most Valuable

A real option is most valuable — and staging is most worthwhile — under these conditions:

Condition	Why it matters
High cost of failure	If the downside of being wrong is large, early learning saves a lot.
High probability of failure	When most companies fail, experiments are cheap on an expected-value basis.
Experiment is cheap vs. total investment	If Stage 1 costs 1% of total capital, the savings from abandoning are large.
Experiment yields accurate signal	Staging is less useful if the experiment doesn't clearly distinguish success from failure.
Less pressure to grow fast	If competitive dynamics force you to deploy capital immediately, staging is harder to execute.

Costs and risks of staging

Experiments too expensive	If Stage 1 costs nearly as much as the full bet, the option has little value.
Experiments take too long	Long timelines may exceed fund life or allow competitors to get ahead.
Perpetual experimentation	Danger of never assimilating information — companies that keep pivoting without deciding.
Coordination frictions	Each new round requires new investors; their beliefs about the company matter as much as yours.

Coordination frictions — the underappreciated risk

When deciding whether to fund a bridge round, the right question is not just 'do I believe in this company?' — it is 'will the *next* investor believe in it?' If Series B investors won't come in, your bridge is a bridge to nowhere. VC investing is therefore pro-cyclical even for contrarians: rational investors anticipate market sentiment in future rounds, making it hard to invest against the crowd even when fundamentals justify it.

6. Fast Ion Battery — Case Analysis

Background

Fast Ion Battery is a cleantech startup developing a revolutionary energy storage technology, founded in May 2008. The company raised a tranchied Series A: £4M in May 2008 (Series A1) and £6M in December 2009 (Series A2). By December 2011, the company has burned through its capital, missed milestones, and replaced its CEO. Bluelock Ventures has just pulled out. WSC's John Davidson must decide whether to fund a £5M bridge round.

The Evaluative Framework: People / Opportunity / Context

Dimension	Assessment for Fast Ion
People	Founder showed poor market focus and management; two promising CEO candidates identified. Key question: is the problem the technology or the team?
Opportunity	Technology has held up and is promising. Pivoted to EV batteries — verbal customer interest but no signed contracts. Large potential market (grid-scale storage + EVs) but long timeline to realise.
Context	Terrible: 2011 cleantech investment is collapsing. Bluelock — a large, capable fund — has opted out. The wider VC market is moving to software/internet. Capital intensity and long timelines of cleantech are a structural mismatch with VC fund structures.

Arguments for funding the bridge

- ✓ £5M is small relative to total projected investment — the bridge is another cheap experiment.
- ✓ The problem may be management, not technology; the new CEO candidates look strong.
- ✓ Bridge terms offer a chance to increase ownership at a flat price (same price as A2) ahead of a potential value inflection point.
- ✓ Warrants available to oversubscribing investors sweeten the deal further.
- ✓ Without a bridge, the company dies; it costs nothing extra to at least wait and see.

Arguments against funding the bridge

- ✗ Financing risk: Bluelock opting out sends a powerful negative signal — 'what does Donna know that John doesn't?'
- ✗ WSC is a small £100M fund; cannot fund Fast Ion to cash flow positive alone. Entirely dependent on Series B investors who may not come.
- ✗ They failed to hit milestones over 3.5 years — why will 12 more months be different?
- ✗ Sunk cost bias: John needs a win; this may be clouding his judgment. His partners should scrutinise this.
- ✗ If the bridge doesn't attract a Series B, WSC loses both its original £10M and the new £5M.

The sunk cost trap

The right question is NOT 'I've already put in £10M, so I should protect it.' It is: 'What is the expected value of my total ownership if I invest the additional £5M, compared to the opportunity cost of deploying that capital elsewhere?' Past investments are gone either way — only future cash flows matter.

What does the bridge need to deliver?

To attract a Series B and overcome financing risk, the bridge round must generate concrete proof points within 12 months — not just progress, but signals that will convince an outside investor. Specifically:

- Signed customer contracts or letters of intent (not just verbal interest)
- A credible commercial partnership
- A proven new CEO with a clear go-to-market plan
- Strong customer references and independent technical validation

What actually happened

John became CEO himself and invoked the pay-to-play clause against Bluelock. The company secured further financing, formed a partnership in China, and was gaining traction. Then a Chinese regulatory change restricting capital outflows killed a late-stage PE term sheet at the last moment. The company went bankrupt. Bad luck at the end — not necessarily a bad decision to fund the bridge.

7. Pay-to-Play Mechanics & Cap Table Impact

What is pay-to-play?

A pay-to-play (PTP) provision forces existing investors to participate in the current round at their pro-rata share, or face a severe penalty: their preferred stock converts to common stock at a punitive ratio. In Fast Ion's bridge terms, the ratio is **10:1** — 10 shares of preferred convert to 1 share of common.

Dimension	Detail
Purpose	Prevents free-riding. Stops an investor from opting out of a risky round while still benefiting from the upside if the company succeeds on others' capital.
Effect on cap table	Massively dilutes the non-participating investor. Bluelock's preferred converts to common at 10:1, wiping out nearly all their economic interest and preference protection.
Effect on remaining investors	WSC and Franconia increase their ownership percentage and improve their position in the liquidation waterfall.
Signal to future investors	Cleans up the cap table — no awkward overhang from a large investor who has mentally exited. Series B investors prefer a clean, committed syndicate.

Should John invoke pay-to-play against Donna?

Arguments for invoking PTP:

- ✓ Financially it makes sense — increases WSC's ownership ahead of a potential value inflection.
- ✓ Bluelock is free-riding if they benefit from John's risk without contributing.
- ✓ Cleans the cap table for future investors, making the Series B story simpler.
- ✓ In some markets (e.g. UK biotech) this is standard practice and not seen as aggressive.

Arguments against invoking PTP:

- ✗ VC is a small world. Donna has a stronger track record than John. This burns a relationship.
- ✗ It signals aggression that could affect WSC's ability to co-invest with Bluelock in future deals.
- ✗ Donna pulled out because of genuine fund-level concerns (cleantech thesis failing) — not bad faith.
- ✗ The reputational cost may exceed the financial benefit if John is wrong about Fast Ion's potential.

Cap Table: Bridge Round (with PTP invoked)

Investor	Bridge Shares	Cumul. Ownership	Notes
Founder / Mgmt	5,000,000	33%	Unchanged from A2
WSC (A1+A2+Bridge)	5,083,333	33%	Includes warrants
Franconia (A1+A2+Bridge)	5,083,333	33%	Includes warrants
Bluelock (post-PTP)	200,000	1%	10:1 conversion; preferred → common
Total	15,366,667	100%	

8. Liquidation Waterfall — GreenTech Example

Worked example from the problem set. Used here to illustrate how liquidation preferences interact with participating preferred stock.

Setup

Seed round	VC Orion invests £4M at £16M pre-money. 10M shares, split 8M founder / 2M Orion.
Series A	VC Zenith invests £12M at £48M pre-money. 10% option pool created on fully-diluted post-money basis. Zenith receives participating convertible preferred with 2x liquidation preference.
Exit	£80M

Series A cap table (post-money = £60M)

Shareholder	Shares	Ownership	Value at Exit
Founder	8,000,000	56%	—
VC Orion	2,000,000	14%	—
Option Pool	1,428,571	10%	—
VC Zenith	2,857,143	20%	—
Total	14,285,714	100%	£60M post-money

Key calculation: option pool shares

The option pool must equal 10% of all fully-diluted post-money shares. This requires solving simultaneously for the pool size and share price:

Let N = total post-money shares, S_O = option pool shares, S_Z = Zenith shares

$$S_O = 0.10 \times N \text{ and } N = 10M + S_O + S_Z$$

$$\text{Price } P = £48M / (10M + S_O) \text{ and } S_Z = £12M / P = 0.25 \times (10M + S_O)$$

$$\text{Solving: } S_O = 1,428,571 \mid S_Z = 2,857,143 \mid P = £4.20/\text{share}$$

Exit waterfall at £80M (participating preferred)

Zenith's preferred is **participating** — she gets her 2x preference back *and* then participates pro-rata in the remainder. This is the most investor-friendly structure.

Step	Recipient	Amount	Basis
1 — Liquidation preference	VC Zenith	£24M	2x × £12M invested
2 — Remaining pool	All shareholders	£56M	£80M – £24M
Founder (56%)		£31.4M	56% × £56M

VC Orion (14%)		£7.8M	$14\% \times £56M$
VC Zenith (20%)		£11.2M	$20\% \times £56M$
Option Pool (10%)		£5.6M	$10\% \times £56M$
Total — VC Zenith	VC Zenith	£35.2M	$£24M + £11.2M$
Total	All	£80M	

Participating vs. Non-participating preferred

Non-participating: Zenith takes preference (£24M) OR converts to common and takes pro-rata share ($20\% \times £80M = £16M$) — whichever is higher. At £80M exit she takes £24M and does not share in the remainder. Remaining £56M split among all *except* Zenith.

Participating (this case): Zenith takes £24M preference AND shares pro-rata in the remaining £56M. Total = £35.2M. Always better for the investor; always worse for founders and other shareholders.

9. Key Takeaways

1

VC returns are highly skewed and persistent

A few hits drive the entire asset class. Top-quartile managers stay top-quartile — unlike public markets. Picking the right VC fund matters enormously.

2

Multi-stage financing = buying options

Each round buys time for an experiment. The value comes from the option to abandon, not from the experiment itself. Staging reduces exposure to losers.

3

Staged financing panel example

Same 1% unconditional success probability → Panel A (all-or-nothing) is NPV negative, Panel B (staged) has £8.9M option value. The difference is when you commit capital.

4

Real options are most valuable when failure is costly, likely, and experiments are cheap

Cleantech fails this test in part — the 'experiments' are expensive and long, reducing the option value of staging.

5

Coordination frictions are as important as fundamentals

Ask not just 'do I believe in this company?' but 'will the next investor?' If future capital won't come, your bridge round is worthless.

6

Pay-to-play prevents free-riding

Forces all investors to commit or face punitive dilution. Good for cap table hygiene and future investor optics, but costly to relationships in a small industry.

7

Liquidation waterfalls — know the order

Preference first, then pro-rata for the remainder (if participating). Non-participating preferred holders choose between preference and conversion.

8

Option pool shuffling dilutes existing holders

Creating the pool before pricing the round reduces the share price, diluting founders and earlier investors — not the new investor.